# Model 8: Identification with Reversed Information Structure

**Note on rigor.** The identification proof in this document is more detailed and formally argued than is standard in the empirical IO literature—most papers in this area assert identification via heuristic counting arguments or brief sketches. That said, a few elements are not fully nailed down:

- One footnote was beyond my understanding, I flagged it so that I ask someone else to check it. [footnote 6]
- footnoe 7 about the shape columns is something I could just not check given its complexity

but both things make intuitive sense.

In the baseline model where $\mathbf{x}$ is public information (both banks observe it), bank 1—the incumbent—knows both the borrower's default probability $h$ and the competitive environment $r_2^*(\mathbf{x})$ it faces. This creates an identification problem: when $\mathbf{x}$ is unobserved by the econometrician, bank 1's equilibrium rate $r_1 = \sigma^*(h; r_2^*(\mathbf{x}))$ depends on two latent variables, and the mapping from observed rates to borrower types is many-to-one.

This document studies an alternative information structure in which bank 1 observes only $h$—not the covariates $\mathbf{x}$—while bank 2 continues to observe $\mathbf{x}$. The economic logic: the incumbent's informational advantage is about credit quality (learned through the lending relationship), while the entrant's advantage is about the competitive environment (what the borrower looks like to outside lenders, what offers they receive, etc.). Bank 1 knows how risky the borrower is, but not how attractive the borrower is to competitors.

The key result: since bank 1 cannot condition on $\mathbf{x}$, its equilibrium strategy $\sigma^*(h)$ depends only on $h$, restoring a one-to-one mapping from bank-1 rates to borrower types. This dramatically simplifies identification.

Section 1 sets up the model. Section 2 derives the equilibrium. Section 3 analyzes identification. Section 4 derives the likelihood. Section 5 compares this model with the public-$\mathbf{x}$ baseline. Appendix A extends the identification analysis to the case of continuous covariates.

## 1 Model

Two banks compete for a borrower. The borrower has default probability $h \in [\underline{h}, \bar{h}]$ and covariates $\mathbf{x} \in \mathcal{X}$. The covariates and default probability are related: $h$ and $\mathbf{x}$ are jointly distributed with conditional density $f(h|\mathbf{x})$, so that knowing $h$ is informative about $\mathbf{x}$ and vice versa.

*Assumption* 1 (Finite covariates). The covariates take finitely many values: $\mathbf{x} \in \{x_1, \ldots, x_K\}$, with population probabilities $p_j = \Pr(\mathbf{x} = x_j) > 0$, $\sum_{j=1}^K p_j = 1$. Bank 2's equilibrium strategy assigns a rate $r_{2,j}^* = r_2^*(x_j)$ to each covariate value, so that the distribution of entrant rates $G$ is discrete on at most $K$ support points.

This assumption is natural in practice: credit scores are grouped into bins, income into brackets, geography into discrete markets. It is also maintained in the public-$\mathbf{x}$ simulations ($K = 5$).

The key consequence is that $G$ is finite-dimensional—fully described by the $K$ support points $\{r^*_{2,1}, \ldots, r^*_{2,K}\}$ and the $K$ probabilities $\{p_1, \ldots, p_K\}$—eliminating the need for any parametric assumption on the mixing distribution.

## 1.1 Information Structure

| | **Bank 1 (incumbent)** | **Bank 2 (entrant)** |
|---|---|---|
| Default probability $h$ | Observed | Not observed |
| Covariates $\mathbf{x}$ | **Not observed** | Observed |
| Distribution $F(h\|\mathbf{x})$ | Knows $F(h\|\mathbf{x})$ and $p_j$ | Knows $F(h\|\mathbf{x})$ |

Bank 1 knows the borrower's true risk from the lending relationship, but does not observe the covariates $\mathbf{x}$ that determine the competitive environment. However, since $h$ and $\mathbf{x}$ are correlated, bank 1 can form a posterior belief $\pi_j(h) \equiv \Pr(\mathbf{x} = x_j \mid h)$ about the borrower's covariates using Bayes' rule. Bank 2 observes $\mathbf{x}$ and uses it to set its rate, but does not know $h$.

## 1.2 Strategies and Demand

Bank 1 sets $r_1 = \sigma(h)$, a function of $h$ alone. Bank 2 sets $r_2 = \rho(\mathbf{x})$, a function of $\mathbf{x}$. The borrower observes both offers and chooses according to a logit:

$$s(r_1, r_2) = \frac{1}{1 + \exp\big(\alpha(r_1 - r_2) - \lambda\big)} \tag{1}$$

where $\alpha > 0$ is rate sensitivity and $\lambda$ captures switching costs (bank-1 advantage when $\lambda > 0$).

## 1.3 Payoffs

Bank $j$'s expected profit from lending to a borrower with type $h$ at rate $r_j$ is $(r_j(1-h) - 1)$, with break-even rate $1/(1-h)$. Bank 1 knows $h$; bank 2 faces the distribution $F(h|\mathbf{x})$.

## 1.4 Data

For each borrower $i = 1, \ldots, N$, the econometrician observes:

| **Variable** | **Description** | **Values** |
|---|---|---|
| $r_i^{\text{win}}$ | Interest rate charged by the chosen bank | $\mathbb{R}_+$ |
| $W_i$ | Identity of the chosen bank | $\{1, 2\}$ |
| $D_i$ | Default indicator | $\{0, 1\}$ |

As in the public-$\mathbf{x}$ case, the econometrician does <u>not</u> observe $\mathbf{x}$. The question is what can be recovered from $(r_i^{\text{win}}, W_i, D_i)$.

# 2 Equilibrium

## 2.1 Bank 1's Problem

Bank 1 knows $h$ but not $\mathbf{x}$. It does not know which $r_2 = r_2^*(x_j)$ this particular borrower will face. However, since $h$ and $\mathbf{x}$ are correlated, observing $h$ is informative about $\mathbf{x}$. By Bayes' rule, bank 1's posterior probability that the borrower has covariates $x_j$ is:

$$\pi_j(h) \equiv \Pr(\mathbf{x} = x_j \mid h) = \frac{p_j\, f(h \mid x_j)}{\sum_{k=1}^{K} p_k\, f(h \mid x_k)} \tag{2}$$

where $f(h|x_j)$ is the conditional density of $h$ given $\mathbf{x} = x_j$.

Bank 1 maximizes expected profit using this posterior:

$$\max_{r_1} \ \left(r_1 - \tfrac{1}{1-h}\right) \cdot \bar{S}(r_1;\, h) \tag{3}$$

where

$$\bar{S}(r_1;\, h) \equiv \sum_{j=1}^{K} \pi_j(h)\, s(r_1,\, r_{2,j}^*) \tag{4}$$

is the <u>expected market share</u>, averaging over the $K$ competitive environments weighted by bank 1's posterior belief about which environment applies. Note that $\bar{S}$ now depends on $h$ through two channels: directly through the logit share $s$ (via the rate $r_1 = \sigma^*(h)$), and indirectly through the posterior weights $\pi_j(h)$ (riskier borrowers are more likely to have certain covariates).

The first-order condition is:

$$\bar{S}(\sigma^*(h);\, h) + \left(\sigma^*(h) - \tfrac{1}{1-h}\right) \cdot \bar{S}_1(\sigma^*(h);\, h) = 0 \tag{5}$$

where $\bar{S}_1$ denotes the partial derivative with respect to the first argument. Since $\bar{S}_1(r_1;\, h) = -\alpha\, \bar{V}(r_1;\, h)$ where $\bar{V}(r_1;\, h) \equiv \sum_{j=1}^{K} \pi_j(h)\, s(r_1, r_{2,j}^*)(1 - s(r_1, r_{2,j}^*))$, this gives:

$$\boxed{\sigma^*(h) = \frac{1}{1-h} + \frac{1}{\alpha} \cdot \frac{\bar{S}(\sigma^*(h);\, h)}{\bar{V}(\sigma^*(h);\, h)}} \tag{6}$$

The equilibrium rate equals the break-even rate $1/(1-h)$ plus a markup. The markup is $(1/\alpha)$ times the ratio of expected share to expected share-variance, both evaluated at bank 1's posterior beliefs $\pi(h)$.

*Remark* 2 (Comparison with the public-$\mathbf{x}$ case). When bank 1 observes $\mathbf{x}$ (and hence knows $r_2$), the markup is $1/(\alpha(1 - s(\sigma^*(h; r_2), r_2)))$—the inverse semi-elasticity evaluated at the specific $r_2$. Here, bank 1 optimizes against the <u>posterior distribution</u> of $r_2$ given $h$, with weights $\pi_j(h)$ that vary across borrower types. By Jensen's inequality, $\bar{S}/\bar{V} \neq 1/(1 - \bar{S})$ in general: bank 1 cannot replicate the informed strategy by using the "average" competitive environment.

## 2.2 Bank 2's Problem

Bank 2 observes $\mathbf{x}$ and knows the conditional distribution $F(h|\mathbf{x})$. It also knows bank 1's equilibrium strategy $\sigma^*(\cdot)$ (which is common knowledge). Bank 2 maximizes:

$$\max_{r_2} \int \big(r_2(1-h)-1\big) \cdot \big(1 - s(\sigma^*(h), r_2)\big)\, dF(h|\mathbf{x}) \tag{7}$$

The first-order condition gives:

$$\boxed{r_2^*(\mathbf{x}) = \frac{\int (1 - s(\sigma^*(h), r_2^*))(1-h)\, dF(h|\mathbf{x})}{\alpha \int s(\sigma^*(h), r_2^*)(1 - s(\sigma^*(h), r_2^*))(1-h)\, dF(h|\mathbf{x})} + \frac{\int s(\sigma^*(h), r_2^*)(1 - s(\sigma^*(h), r_2^*))\, dF(h|\mathbf{x})}{\int s(\sigma^*(h), r_2^*)(1 - s(\sigma^*(h), r_2^*))(1-h)\, dF(h|\mathbf{x})}} \tag{8}$$

The entrant's rate depends on $\mathbf{x}$ through the conditional type distribution $F(h|\mathbf{x})$. Different $\mathbf{x}$ generate different competitive environments, producing a distribution of $r_2^*$ values across borrowers.

## 2.3 The One-to-One Result

**Proposition 3** (One-to-one pricing)**.** *Under standard regularity conditions (log-concavity of demand), the equilibrium bank-1 strategy $\sigma^* : [\underline{h}, \bar{h}] \to \mathbb{R}_+$ is strictly increasing in $h$.*

*Proof.* The bank-1 FOC (6) defines $\sigma^*(h)$ implicitly as the solution to $\sigma = c(h) + M(\sigma, h)$, where $c(h) = 1/(1-h)$ and $M(\sigma, h) = \bar{S}(\sigma; h)/(\alpha\, \bar{V}(\sigma; h))$ is the markup. The type $h$ enters the markup through two channels: (i) the logit shares $s$ (via the rate $\sigma$), and (ii) the posterior weights $\pi_j(h)$. By the implicit function theorem:

$$\frac{d\sigma^*}{dh} = \frac{c'(h) + \partial M/\partial h}{1 - \partial M/\partial \sigma}$$

The term $c'(h) = 1/(1-h)^2 > 0$. For the logit demand, $\partial M/\partial \sigma < 1$ at the equilibrium (this is the second-order condition), so the denominator is positive. The term $\partial M/\partial h$ captures how the posterior belief $\pi(h)$ shifts toward different competitive environments as $h$ changes. Under standard regularity conditions (the shift in posterior weights does not overwhelm the cost channel), the numerator is positive. Hence $d\sigma^*/dh > 0$. □

Since $\sigma^*$ is strictly increasing, its inverse $\tau(r_1) \equiv (\sigma^*)^{-1}(r_1)$ is well-defined and strictly increasing.

**Definition 4** (Type-recovery function)**.** $\tau : [\sigma^*(\underline{h}), \sigma^*(\bar{h})] \to [\underline{h}, \bar{h}]$ is defined by $\tau(\sigma^*(h)) = h$ for all $h$.

*Remark* 5 (Why this differs from the public-$\mathbf{x}$ case)*.* When $\mathbf{x}$ is public information, bank 1 observes $\mathbf{x}$, so its strategy $\sigma^*(h; \mathbf{x})$ maps the <u>pair</u> $(h, \mathbf{x})$ to $r_1$. The same $r_1$ can arise from different $(h, \mathbf{x})$ combinations, making the mapping many-to-one and breaking type recovery. Here, $\sigma^*(h)$ depends on $h$ alone, so each $r_1$ corresponds to exactly one $h$. The inversion is possible regardless of whether $\mathbf{x}$ is observed.

# 3 Identification

## 3.1 Type Recovery

The central identification result follows immediately from Proposition 3.

**Proposition 6** (Type recovery)**.** *Among bank-1 winners, the conditional default rate identifies the borrower's type:*

$$E[D_i \mid r_i^{win} = r_1,\, W_i = 1] = \tau(r_1) \tag{9}$$

*for all $r_1$ in the support of bank-1 winning rates.*

*Proof.* Since $\sigma^*$ is one-to-one (Proposition 3), conditioning on $r_1 = \sigma^*(h)$ uniquely determines $h = \tau(r_1)$. Default is $D_i \sim \text{Bernoulli}(h_i)$, independent of the bank identity $W_i$ conditional on $h_i$ (default is a borrower characteristic, not a bank characteristic).[1] Therefore:

$$E[D_i \mid r_1, W_i = 1] = E[D_i \mid h_i = \tau(r_1)] = \tau(r_1)$$

Crucially, this holds for <u>all</u> bank-1 winners at rate $r_1$, regardless of their unobserved $\mathbf{x}$. Different borrowers at the same $r_1$ have different $\mathbf{x}$ (and hence face different $r_2^*$), but they all have the same $h = \tau(r_1)$ and therefore the same default probability. □

*Remark* 7 (Testable implication). Type recovery implies that among bank-1 winners at rate $r_1$, defaults are i.i.d. Bernoulli($\tau(r_1)$). Therefore:

$$\text{Var}(D_i \mid r_1, W_i = 1) = \tau(r_1)\big(1 - \tau(r_1)\big) \tag{10}$$

This is the Bernoulli variance—there is no "excess variance" from type heterogeneity. In the public-$\mathbf{x}$ case (when the econometrician does not observe $\mathbf{x}$), borrowers at the same $r_1$ can have different types, so $\text{Var}(D_i \mid r_1, W_i = 1) > h(1-h)$ due to the mixture. This provides a direct specification test to distinguish the two information structures.

## 3.2 What Is Identified Nonparametrically

**Proposition 8** (Nonparametrically identified objects)**.** *From the joint distribution of $(r_i^{win}, W_i, D_i)$, the following are nonparametrically identified:*

*(a)* ***The type-recovery function*** $\tau(r_1)$**.** *Equivalently, the full strategy $\sigma^*(h)$ is traced out by regression of $D_i$ on $r_i^{win}$ among bank-1 winners.*

[1] **Note:** The independence $D_i \perp W_i \mid h_i$ is used here but is not stated as a formal assumption. It holds by construction in this model because the default probability $h$ is a primitive of the borrower and the payoff specification $(r_j(1-h) - 1)$ treats $h$ as the same regardless of which bank lends. However, this rules out settings where the choice of bank affects default risk—for instance, if banks differ in loan amounts, maturities, or covenants that influence the borrower's ability to repay, or if bank 2 attracts borrowers by offering looser terms that increase default probability. In Chilean consumer credit markets, standardized consumer loan products may make this less of a concern, but it should be acknowledged as a maintained assumption, particularly since the entire type-recovery result depends on it. Consider adding it as a formal assumption (e.g., "$D_i \perp W_i \mid h_i$: default probability depends only on the borrower's type, not on the identity of the chosen bank").

(b) ***The markup function*** $m(h) = \sigma^*(h) - 1/(1-h)$***.*** *Since* $\sigma^*(h)$ *and* $1/(1-h)$ *are both known functions of* $h$ *(the former from type recovery), the markup—the pure profit bank 1 earns above break-even—is identified for every type.*

(c) ***The selected distribution of*** $r_2^*$ ***among bank-2 winners.*** *Among bank-2 winners,* $r_i^{win} = r_2^*(\mathbf{x}_i)$*, so the empirical distribution of winning rates directly reveals the selection-weighted distribution of entrant rates.*

(d) ***The conditional default rate among bank-2 winners,*** $E[D_i \mid r_2, W_i = 2]$***.*** *This is the adverse-selection-weighted mean type among borrowers who switch to the entrant at rate* $r_2$*.*

(e) ***The selected distribution of types*** $h$ ***among bank-1 winners.*** *From type recovery,* $h_i = \tau(r_i^{win})$ *for each bank-1 winner, so the empirical distribution of types among stayers is identified.*

(f) ***The aggregate switching probability*** $\Pr(W_i = 2)$***.***

*Proof.* Items (c), (d), (f) are features of the observable joint distribution, requiring no model structure. Items (a), (b), (e) follow from Proposition 6 and the known parametric form of the break-even rate $1/(1-h)$. □

The critical difference from the public-$\mathbf{x}$ case: items (a), (b), and (e) are <u>not</u> identified when $\mathbf{x}$ is public but unobserved by the econometrician, because there, the many-to-one mapping from $(h, \mathbf{x})$ to $r_1$ prevents type recovery. Here, they follow directly.

## 3.3 Full Identification

Proposition 8 identifies the bank-1 strategy $\sigma^*(h)$, the markup $m(h)$, and the support of bank-2 rates nonparametrically. The remaining objects—the demand parameters $(\alpha, \lambda)$, the population probabilities $(p_1, \ldots, p_K)$, and the conditional type distributions $f(h|x_j)$—are identified from the equilibrium first-order conditions together with bank-2 default rates. For $K = 2$, no parametric assumption on $F(h|\mathbf{x})$ is needed. For $K \geq 3$, a finite-dimensional parametric family for $f(h|x_j)$ is required.

The full parameter vector is $\theta = (\alpha, \lambda, f(\cdot|x_1), \ldots, f(\cdot|x_K), p_1, \ldots, p_K)$, where the $r_{2,j}^*$ are treated as observed. For $K \geq 3$ (or for practical estimation when $K = 2$), one parametrizes $f(h|x_j) = f(h;\, \theta_j)$ with a $P$-parameter family, reducing the unknowns to $(\alpha, \lambda, \theta_1, \ldots, \theta_K, p_1, \ldots, p_K)$.

**Proposition 9** (Identification)**.** *Under Assumption 1, the demand parameters* $(\alpha, \lambda)$*, population probabilities* $(p_1, \ldots, p_K)$*, and conditional type distributions* $f(h|x_j)$ *for* $j = 1, \ldots, K$ *are identified from the joint distribution of* $(r_i^{win}, W_i, D_i)$ *if:*

*(C1)* $K \geq 2$ *(at least two distinct covariate values) and bank-2 winners are observed at each* $r_{2,j}^*$*.*

*(C2)* $\alpha > 0$ *and* $\Pr(W_i = 2) > 0$*.*

*(C3) When* $K \geq 3$*:* $f(h|x_j) = f(h;\, \theta_j)$ *for a* $P$*-parameter family* $\{f(\cdot;\theta) : \theta \in \Theta \subseteq \mathbb{R}^P\}$ *satisfying regularity conditions (P1)–(P2) stated in the proof.*

*When* $K = 2$*, condition (C3) is not needed:* $f(h|x_j)$ *is nonparametrically identified.*

*Proof.* The proof proceeds in stages, exploiting the separation between bank-1 and bank-2 data that type recovery makes possible.

**Stage 1: Recovery of $\sigma^*(h)$ and $m(h)$ (nonparametric).**

From Proposition 6, the regression $E[D_i \mid r_1, W_i = 1]$ traces the type-recovery function $\tau(r_1)$, hence $\sigma^*(h) = \tau^{-1}(h)$. The markup $m(h) = \sigma^*(h) - 1/(1-h)$ follows. This stage uses only bank-1 data and requires no parametric assumptions.

**Stage 2: Recovery of the support points $\{r^*_{2,j}\}$ (nonparametric).**

Among bank-2 winners, $r_i^{\text{win}} = r^*_{2,j}$ for some $j$. Since $\mathbf{x}$ takes $K$ distinct values, bank-2 winning rates take at most $K$ distinct values. These are directly observed from the data.

**Stage 3: Identification mechanism.**

At each $h$, the FOC and the observed density of bank-1 winner types together provide <u>two</u> restrictions on the $K$ unknowns $\gamma_j(h) \equiv p_j\, f(h|x_j)$. When $K = 2$, this exactly pins down $\gamma_1(h)$ and $\gamma_2(h)$ pointwise, giving nonparametric identification. When $K \geq 3$, the pointwise system is underdetermined and a finite-dimensional restriction on $f$ is needed.

**The two pointwise restrictions.** Fix candidate values $(\alpha, \lambda)$. At each $h$, define $s_j(h) \equiv s(\sigma^*(h), r^*_{2,j};\, \alpha, \lambda)$ (determined from Stages 1–2 for a given candidate).

<u>Restriction 1 (FOC).</u> The bank-1 first-order condition requires:

$$\sum_{j=1}^{K} \gamma_j(h)\, s_j(h) \Big[1 - \alpha\, m(h)\, \big(1 - s_j(h)\big)\Big] = 0 \tag{R1}$$

This is one linear equation in $(\gamma_1, \ldots, \gamma_K)$ with known, $h$-varying coefficients.

<u>Restriction 2 (bank-1 type density).</u> Among bank-1 winners, type recovery gives $h_i = \tau(r_i^{\text{win}})$, so the density $f(h \mid W_i = 1)$ is directly observed from data. Since $\Pr(W_i = 1 \mid h) = \sum_j \gamma_j(h)\, s_j(h)/f(h)$:

$$\sum_{j=1}^{K} \gamma_j(h)\, s_j(h) = \Pr(W_i = 1) \cdot f(h \mid W_i = 1) \equiv C(h) \tag{R2}$$

where $C(h)$ is directly computable from data (the type density among bank-1 winners, rescaled by the bank-1 win probability).

Together, (R1) and (R2) are two equations in the $K$ unknowns $\gamma_1(h), \ldots, \gamma_K(h)$ at each $h$.

**Case $K = 2$: nonparametric identification.**

The system (R1)–(R2) is $2 \times 2$ in $(\gamma_1(h), \gamma_2(h))$. Since $r^*_{2,1} \neq r^*_{2,2}$ (condition C1), the shares $s_1(h) \neq s_2(h)$ and the coefficient vectors in (R1) and (R2) are linearly independent for generic $h$. Solving gives:

$$\gamma_j(h;\, \alpha, \lambda) = p_j\, f(h|x_j;\, \alpha, \lambda) \quad \text{determined at every } h$$

This is a known, explicit function of the candidate $(\alpha, \lambda)$ and $h$.

<u>Recovering $p_j$.</u> The integrability constraints $\int f(h|x_j)\, dh = 1$ give:

$$p_j = \int \gamma_j(h;\, \alpha, \lambda)\, dh, \quad j = 1, 2$$

This determines $p_1$ (and $p_2 = 1 - p_1$) as a function of $(\alpha, \lambda)$.

Pinning down $(\alpha, \lambda)$. The bank-2 conditional default rates provide two moment conditions:

$$E[D_i \mid r^*_{2,j},\, W_i = 2] = \frac{\int h\,(1 - s_j)\, f(h|x_j)\, dh}{\int (1 - s_j)\, f(h|x_j)\, dh}, \quad j = 1, 2$$

where everything on the right depends on $(\alpha, \lambda)$ through the steps above. Two equations in two unknowns: identified. The aggregate switching rate $\Pr(W_i = 2) = \sum_j p_j \int (1 - s_j)\, f(h|x_j)\, dh$ provides an additional overidentifying restriction.

Restriction count $(K = 2)$. At each $h$: 2 equations, 2 unknowns—exactly identified pointwise, no parametric assumption needed. Globally: 2 integrability constraints determine $p_1$ (1 free parameter, 1 overidentifying restriction). Two bank-2 default rates plus the switching rate give 3 equations for $(\alpha, \lambda)$ (2 unknowns, 1 overidentifying restriction). Total excess restrictions: **2**.

**Case $K \geq 3$: parametric restriction needed.**

The system (R1)–(R2) provides 2 equations in $K$ unknowns at each $h$—underdetermined for $K \geq 3$. The system has $K - 2$ free parameters per $h$, so infinitely many density profiles $f(h|x_j)$ are consistent with the FOC and the observed type density. Nonparametric identification of $f(h|x_j)$ fails when $K \geq 3$.

Resolution: parametric identification. Impose condition (C3): let $f(h|x_j) = f(h;\, \theta_j)$ for a $P$-parameter family $\{f(\cdot;\theta) : \theta \in \Theta \subseteq \mathbb{R}^P\}$ satisfying:

(P1) Identifiability: $f(\cdot;\theta) = f(\cdot;\theta')$ a.e. implies $\theta = \theta'$.

(P2) Distinguishability: for any $K$ distinct vectors $\theta_1, \ldots, \theta_K \in \Theta$, the densities $f(\cdot;\theta_1), \ldots, f(\cdot;\theta_K)$ are linearly independent on $[\underline{h}, \bar{h}]$.[2]

Both conditions hold for Beta, log-normal, Gamma, and truncated normal families. The finite-dimensional parameter vector is $\psi = (\alpha, \lambda, \theta_1, \ldots, \theta_K, p_1, \ldots, p_{K-1}) \in \mathbb{R}^{KP+K+1}$, where $p_K = 1 - \sum_{j<K} p_j$. We establish identification in two steps.[3]

Step A: $(p_j, \theta_j)_{j=1}^K$ are identified given $(\alpha, \lambda)$.

Fix candidate values $(\alpha, \lambda)$. The logit shares $s_j(h) = s(\sigma^*(h), r^*_{2,j};\, \alpha, \lambda)$ are then known, strictly positive functions on the interior of $[\underline{h}, \bar{h}]$. Under the parametric assumption, $\gamma_j(h) = p_j\, f(h;\theta_j)$,

[2] Linear independence of densities means there is no set of constants $c_1, \ldots, c_K$, not all zero, such that $\sum_{j=1}^K c_j\, f(h;\theta_j) = 0$ for almost every $h \in [\underline{h}, \bar{h}]$. This rules out the pathological case where two covariate groups produce proportional type distributions (e.g., $f(h;\theta_2) = a\, f(h;\theta_1)$ for some constant $a$), which would make the groups observationally indistinguishable in the (R1)–(R2) system. The condition holds generically for standard parametric families: for instance, two Beta densities $\text{Beta}(a_1, b_1)$ and $\text{Beta}(a_2, b_2)$ with $(a_1, b_1) \neq (a_2, b_2)$ are linearly independent because their ratio $f(h;\theta_1)/f(h;\theta_2)$ is a non-constant function of $h$ (it is a power function $h^{a_1 - a_2}(1-h)^{b_1 - b_2}$ times a constant).

[3] The full vector $\psi$ includes $\alpha$ and $\lambda$, but they do not appear in the Jacobian $J$ of Step A because Step A conditions on a fixed candidate $(\alpha, \lambda)$. For given $(\alpha, \lambda)$, the logit shares $s_j(h) = s(\sigma^*(h), r^*_{2,j};\, \alpha, \lambda)$ are known functions of $h$ (since $\sigma^*(h)$ and $r^*_{2,j}$ come from Stages 1–2). Therefore, only the density parameters $(\theta_1, \ldots, \theta_K)$ and the mixing weights $(p_1, \ldots, p_{K-1})$ remain as unknowns in (R2′), and the Jacobian $J$ is taken with respect to these $D = KP + K - 1$ variables alone. The parameters $(\alpha, \lambda)$ are then identified separately in Step B, where they enter through both $s_j(h;\alpha,\lambda)$ and the functions $\theta_j(\alpha,\lambda)$, $p_j(\alpha,\lambda)$ recovered in Step A.

so restriction (R2) becomes:

$$\sum_{j=1}^{K} p_j\, f(h;\theta_j)\, s_j(h) \;=\; C(h) \qquad \text{for all } h \in [\underline{h},\, \bar{h}] \tag{R2$'$}$$

This is a continuum of equations in the $D \equiv KP + K - 1$ unknowns $(\theta_1, \ldots, \theta_K,\, p_1, \ldots, p_{K-1})$.

Local identification. Select $D$ distinct evaluation points $h_1, \ldots, h_D$ and define $\Phi : \mathbb{R}^D \to \mathbb{R}^D$ by $\Phi_d(\psi_f) = \sum_{j=1}^{K} p_j\, f(h_d;\theta_j)\, s_j(h_d)$, which is simply the model predicted density value that an individual with default probability $h_d$ chooses the incumbent. [4] The Jacobian $J \equiv \partial\Phi/\partial\psi_f$ has two blocks of columns:

- Weight columns ($\partial/\partial p_j$, $j = 1, \ldots, K-1$): entry $d$ is $f(h_d;\theta_j)\, s_j(h_d) - f(h_d;\theta_K)\, s_K(h_d)$.[5]
- Shape columns ($\partial/\partial\theta_{j\ell}$, $j = 1, \ldots, K$, $\ell = 1, \ldots, P$): entry $d$ is $p_j\, (\partial f/\partial\theta_{j\ell})(h_d;\theta_j)\, s_j(h_d)$.

Under (P1)–(P2), the weight columns are spanned by linearly independent functions (since $f(\cdot;\theta_j)\, s_j \neq f(\cdot;\theta_k)\, s_k$ for $j \neq k$ when $r^*_{2,j} \neq r^*_{2,k}$),[6] and the shape columns add linearly independent score directions for each component.[7] For generically chosen evaluation points, $J$ has full rank $D$, so by

[4] The map $\Phi$ is constructed by evaluating the left-hand side of (R2$'$) at $D$ specific type values. The continuum restriction (R2$'$) requires $\sum_j p_j\, f(h;\theta_j)\, s_j(h) = C(h)$ for every $h$, which is an infinite-dimensional condition. To check whether the finite-dimensional parameter vector $\psi_f = (\theta_1, \ldots, \theta_K, p_1, \ldots, p_{K-1})$ is pinned down, we discretize: pick $D = KP + K - 1$ evaluation points $h_1, \ldots, h_D$ and ask whether the system $\Phi_d(\psi_f) = C(h_d)$ for $d = 1, \ldots, D$ has a locally unique solution. Each component $\Phi_d$ is simply the model's predicted value of the bank-1 stayer density at $h_d$ (the left-hand side of (R2$'$) evaluated at $h = h_d$), and the target $C(h_d)$ is the corresponding data object. If the Jacobian $\partial\Phi/\partial\psi_f$ has full rank at the truth, the implicit function theorem guarantees local uniqueness of this $D \times D$ system.

[5] The subtraction arises because $p_K$ depends on $p_j$: $\partial p_K/\partial p_j = -1$. In the vector of parameters $p_K$ is not explicitly included, is implicitly definded as $p_K = 1 - \sum_k p_k$. Intuitively, increasing the population share of group $j$ by $dp_j$ simultaneously decreases the share of the reference group $K$, so the net effect is the difference in the two density-times-share products.

[6] Assumption (P2) gives linear independence of the densities $f(\cdot;\theta_j)$. The weight columns involve the products $f(\cdot;\theta_j)\, s_j - f(\cdot;\theta_K)\, s_K$, so we must verify that multiplying by the logit shares preserves independence. Suppose for contradiction that $\sum_{j=1}^{K-1} c_j\, [f(h;\theta_j)\, s_j(h) - f(h;\theta_K)\, s_K(h)] = 0$ for all $h$, with at least one $c_j \neq 0$. Dividing by $s_K(h) > 0$ gives $\sum_{j<K} c_j\, f(h;\theta_j)\, [s_j(h)/s_K(h)] = (\sum_{j<K} c_j)\, f(h;\theta_K)$. The ratio $s_j(h)/s_K(h) = B_j(B_K + A(h))/(B_K(B_j + A(h)))$, where $B_j = e^{\alpha r^*_{2,j}}$ and $A(h) = e^{\alpha\sigma^*(h)-\lambda}$, is not constant in $h$ . So the equation is a linear combination of the densities with $h$-dependent coefficients, and (P2) cannot be invoked directly. Two observations close the argument. From now on the footnote is more complicated and would have to ask someone like Louis to check it First, all densities $f(\cdot;\theta_j)$ and shares $s_j(\cdot)$ are real-analytic for standard parametric families and the logit (with analytic $\sigma^*$), so by the identity theorem for analytic functions, if an analytic function is zero on a set of positive Lebesgue measure then it is zero everywhere—the linear dependence must hold for all $h$, not just almost all. Second, such an identity would impose a rigid algebraic relationship between the equilibrium pricing schedule $\sigma^*(h)$ (which determines the shares through the logit) and the type densities $f(\cdot;\theta_j)$ (which are governed by the parameters $\theta_1, \ldots, \theta_K$). But $\sigma^*(h)$ is pinned down by the first-order condition (R1), a separate structural restriction that derives from bank 1's profit maximization and does not involve the density parameters $(\theta_j)$. Requiring the FOC-determined pricing $\sigma^*(h)$ and the density family $\{f(\cdot;\theta_j)\}_j$ to jointly satisfy an exact algebraic cancellation $\sum c_j f(\cdot;\theta_j)\, s_j(\cdot;\sigma^*) \equiv 0$ confines the parameter vector to a measure-zero subset of $\mathbb{R}^{KP+K+1}$. For generic parameter values, the products $\{f(\cdot;\theta_j)\, s_j\}_j$ are therefore linearly independent, and so are the $K-1$ weight-column differences $f(\cdot;\theta_j)\, s_j - f(\cdot;\theta_K)\, s_K$.

[7] Each shape column for parameter $(j,\ell)$ has entry $p_j\, (\partial f/\partial\theta_{j\ell})(h_d;\theta_j)\, s_j(h_d)$. Note that perturbing $\theta_{j\ell}$ only affects terms involving group $j$'s density—it has zero effect on $f(\cdot;\theta_k)$ for $k \neq j$. This gives the Jacobian a block-

the implicit function theorem[8] the solution to $\Phi(\psi_f) = (C(h_1), \ldots, C(h_D))^\top$ is locally unique.

Global identification. Any other candidate $\tilde{\psi}_f$ satisfying (R2′) must also match $C(h)$ at the remaining continuum of $h$ values. Since both sides of (R2′) are smooth (in fact analytic for standard parametric families), a $D$-dimensional family of functions that agrees with an observed smooth function on an open interval must be the same function.[9] Hence the locally unique solution is globally unique.

Write the identified objects as $p_j(\alpha, \lambda)$ and $\theta_j(\alpha, \lambda)$.

Step B: $(\alpha, \lambda)$ are identified.

From Step A, each candidate $(\alpha, \lambda)$ determines unique $(p_j(\alpha, \lambda),\, \theta_j(\alpha, \lambda))_{j=1}^K$ via (R2′). Two independent sources of restrictions pin down $(\alpha, \lambda)$.

Bank-2 default rates. Define the observed conditional default rates:

$$\delta_k \;\equiv\; E[D_i \mid r^*_{2,k},\, W_i = 2] \;=\; \frac{\displaystyle\int h\,(1 - s_k(h;\alpha,\lambda))\, f\big(h;\, \theta_k(\alpha,\lambda)\big)\, dh}{\displaystyle\int (1 - s_k(h;\alpha,\lambda))\, f\big(h;\, \theta_k(\alpha,\lambda)\big)\, dh}, \quad k = 1, \ldots, K \qquad (\Delta_k)$$

The left side is observed. On the right, $s_k$ enters through the logit $s = 1/(1+e^{\alpha(r_1-r_2)-\lambda})$, and both $f(\cdot;\theta_k(\alpha,\lambda))$ and $s_k(\cdot;\alpha,\lambda)$ depend on $(\alpha, \lambda)$. These are $K$ nonlinear equations in two unknowns.

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diagonal structure in the shape parameters: the $P$ columns for group $j$ live in the 'subspace' spanned by $f(\cdot;\theta_j)\, s_j$, while the columns for group $k$ live in the subspace spanned by $f(\cdot;\theta_k)\, s_k$. Within group $j$: the $P$ columns are $p_j\, s_j(h)\, \partial_{\theta_{j1}} f,\ \ldots,\ p_j\, s_j(h)\, \partial_{\theta_{jP}} f$. They share the strictly positive factor $p_j\, s_j(h)$, so they are independent if and only if the $P$ partial derivatives $\partial_{\theta_{j1}} f(h;\theta_j),\ \ldots,\ \partial_{\theta_{jP}} f(h;\theta_j)$ are independent as functions of $h$. This is exactly what condition (P1) guarantees: if the density family is identifiable (i.e., different parameters give different densities), then the gradient $\nabla_\theta f$ must have $P$ linearly independent components—otherwise two distinct parameter values could produce the same density by moving in a null direction. Across groups: columns for group $j$ are proportional to $f(\cdot;\theta_j)\, s_j$, and columns for group $k$ are proportional to $f(\cdot;\theta_k)\, s_k$. A linear combination mixing columns from different groups would require $\sum_j \beta_j(h)\, f(h;\theta_j)\, s_j(h) = 0$ for coefficients $\beta_j(h)$ that are themselves linear combinations of the score functions $\partial_\theta \log f$. Since the densities $f(\cdot;\theta_j)$ are linearly independent by (P2) and the scores within each group are independent by (P1), and different groups' contributions are 'modulated' by different density shapes, no cross-group cancellation is possible for generic parameter values. The intuition: perturbing $\theta_j$ moves the model prediction through the shape of $f(\cdot;\theta_j)$, while perturbing $\theta_k$ moves it through the shape of $f(\cdot;\theta_k)$—these are independent directions because the densities themselves are independent.

[8] The result used is the inverse function theorem (a corollary of the implicit function theorem). **Statement:** Let $\Phi : U \subseteq \mathbb{R}^D \to \mathbb{R}^D$ be continuously differentiable on an open set $U$. If $\det(\partial\Phi/\partial\psi_f)\big|_{\psi_f^0} \neq 0$, then there exist open neighborhoods $V \ni \psi_f^0$ and $W \ni \Phi(\psi_f^0)$ such that $\Phi\colon V \to W$ is a bijection with $C^1$ inverse. **Application:** At the true parameters $\psi_f^0$, (R2′) gives $\Phi(\psi_f^0) = (C(h_1), \ldots, C(h_D))^\top$. Full rank of $J$ means $\Phi$ is locally injective, so no other $\psi_f \in V$ produces the same data vector—i.e., $\psi_f$ is locally identified given $(\alpha, \lambda)$. This is the Rothenberg (1971) criterion: a model is locally identified at the truth if the Jacobian of the identifying equations has full rank.

[9] Identity theorem for real-analytic functions: if two real-analytic functions agree on any open interval, they are identical on their entire common domain. Here the argument is as follows. The true parameters $\psi_f^0$ and any alternative $\tilde{\psi}_f$ both satisfy $\sum_j p_j\, f(h;\theta_j)\, s_j(h) = C(h)$ for all $h$ in the evaluation set $\{h_1, \ldots, h_D\}$. But $C(h)$ is observed for every $h \in [\underline{h}, \bar{h}]$, not just at $D$ points. Since $\Phi(\psi_f; h) \equiv \sum_j p_j\, f(h;\theta_j)\, s_j(h)$ is analytic in $h$ for each fixed $\psi_f$ (standard parametric densities and the logit are analytic), and $C(h)$ is also analytic, the difference $\Phi(\psi_f^0;\, h) - \Phi(\tilde{\psi}_f;\, h)$ is analytic and equals zero at the $D$ evaluation points. If $D$ exceeds the number of zeros that a non-trivial analytic function can have on a compact interval (which for analytic functions is at most finite unless the function is identically zero), then the difference must be identically zero for all $h$. That is, $\tilde{\psi}_f$ produces the same function of $h$ as $\psi_f^0$. Combined with local uniqueness (the Jacobian has full rank), this means $\tilde{\psi}_f = \psi_f^0$.

Since the $K$ support points $r^*_{2,k}$ are distinct, different equations weight $(\alpha, \lambda)$ differently in the integral: the logit share $s_k(h)$ shifts by different amounts across $k$, ensuring that the $K$ equations are not redundant. Any two of the $K$ equations generically determine $(\alpha, \lambda)$ uniquely (via the implicit function theorem, since the $2 \times 2$ Jacobian $\partial(\Delta_{k_1}, \Delta_{k_2})/\partial(\alpha, \lambda)$ is nonsingular for generic parameter values when $r^*_{2,k_1} \neq r^*_{2,k_2}$). The remaining $K - 2 \geq 1$ conditions are overidentifying restrictions.

First-order condition. Restriction (R1) must hold for every $h$:

$$\sum_{j=1}^{K} p_j(\alpha, \lambda)\, f\big(h; \theta_j(\alpha, \lambda)\big)\, s_j(h; \alpha, \lambda)\big[1 - \alpha\, m(h)\,(1 - s_j(h; \alpha, \lambda))\big] = 0$$

Since $p_j(\cdot)$, $\theta_j(\cdot)$, and $s_j(\cdot)$ all depend on the candidate $(\alpha, \lambda)$, this is a continuum of additional restrictions that the correct $(\alpha, \lambda)$ must satisfy identically—providing infinitely many overidentifying restrictions beyond the $K$ default-rate conditions.

Restriction count.

| **Source** | **Restrictions** | **Identifies** |
|---|---|---|
| (R2) at continuum of $h$ | $\infty$ (equiv. $KP + K - 1$) | $(\theta_j, p_j)$ given $(\alpha, \lambda)$ |
| $(\Delta_k)$, $k = 1, \ldots, K$ | $K$ | $(\alpha, \lambda)$ + $(K-2)$ overid. |
| (R1) at continuum of $h$ | $\infty$ | Overidentification |
| Switching rate $\Pr(W=2)$ | 1 | Overidentification |

Total unknowns: $KP + K + 1$. Since the continuum restrictions (R1) and (R2) each hold for every $h$ but the parametric structure reduces the unknowns to a finite-dimensional vector, the system is **overidentified** for any $K \geq 3$ and any finite $P$. The effective number of independent identifying restrictions equals the $D = KP + K - 1$ evaluation points needed to pin down $(\theta_j, p_j)$ given $(\alpha, \lambda)$, plus the $K$ bank-2 default-rate conditions and one switching-rate condition, for a total of $KP + 2K$ restrictions against $KP + K + 1$ unknowns—yielding $K - 1$ excess restrictions, with the remaining continuum of $h$ values providing specification tests. □

## 3.4 Why Identification Is Easier Than in the Public-x Case

The fundamental reason is sequential separability:

1. **Stage 1 (type recovery) is model-free.** When $\mathbf{x}$ is public information but unobserved by the econometrician, the econometrician cannot recover $\sigma^*(h)$ without already knowing $(\alpha, \lambda)$—a circular dependence that forces simultaneous estimation of everything. Here, $\sigma^*(h)$ is identified from a simple regression, before any structural estimation begins.

2. **The FOC constrains all structural parameters jointly.** Once $\sigma^*(h)$ and $m(h)$ are known, the FOC constrains $(\alpha, \lambda, f(h|x_j), p_1, \ldots, p_K)$ through the posterior weights $\pi_j(h)$. At each $h$, the FOC and the observed bank-1 type density provide two restrictions on the $K$ unknowns $\gamma_j(h) = p_j f(h|x_j)$. For $K = 2$, this is a $2 \times 2$ system—$f(h|x_j)$ is nonparametrically identified. For $K \geq 3$, nonparametric identification of $f(h|x_j)$ fails and a parametric family is needed, but the system is then overidentified (with $K - 1$ excess restrictions and a continuum

of specification tests). Combined with the $K$ bank-2 default rate restrictions, the system is far more constrained than in the public-$\mathbf{x}$ case, where every restriction involves all parameters simultaneously and the unknown strategy $\sigma^*(h; r_2)$.

3. **Bank-2 data provide independent restrictions on $f(h|x_j)$.** In the public-$\mathbf{x}$ case, the bank-2 default rates constrain $(\alpha, \lambda, f(h|x_j))$ jointly (because $\sigma^*(h; r_2)$ is unknown and depends on all parameters). Here, $\sigma^*(h)$ is already known from bank-1 data, so the bank-2 default rates constrain $f(h|x_j)$ with $(\alpha, \lambda)$ as the only remaining nuisance parameters.

## 3.5 Identification Summary

The table below summarizes the identifying variation for each parameter.

| **Object** | **What identifies it** | **Identifying variation** |
|---|---|---|
| $\sigma^*(h)$, $\tau(r_1)$ | $E[D\|r_1, W = 1]$ (type recovery) | Defaults reveal types; nonparametric regression |
| $m(h) = \sigma^*(h) - \frac{1}{1-h}$ | Directly from $\sigma^*(h)$ | |
| $r^*_{2,j}$ (support of $G$) | Bank-2 winning rates | Directly observed |
| $\alpha$ | Bank-2 default rates | Two moment conditions pin down $(\alpha, \lambda)$ given the $\gamma_j(h)$ functions |
| $\lambda$ | Bank-2 default rates + switching rate | Shifts the logit baseline; pinned down jointly with $\alpha$ |
| $p_j$ (covariate prob's) | Integrability of $\gamma_j(h)$ | $p_j = \int \gamma_j(h)\, dh$; cross-checked against observed $\tilde{p}_j$[10] |
| $f(h\|x_j)$ (type dist'n) | (R1) + (R2) system at each $h$ | $K = 2$: nonparametric (2 eq, 2 unk pointwise). $K \geq 3$: parametric family needed (2 eq, $K$ unk pointwise) |

# 4 Likelihood

## 4.1 Bank-1 Winners ($W_i = 1$)

For a bank-1 winner, $r_i^{\text{win}} = \sigma^*(h_i)$ and type recovery gives $h_i = \tau(r_i^{\text{win}})$. The entrant rate $r_2$ is latent (varies across borrowers through $\mathbf{x}$). Under the finite-$\mathbf{x}$ assumption, the likelihood contribution sums over the $K$ possible competitive environments:

$$\boxed{\mathcal{L}_i^{(1)}(\theta) = |\tau'(r_i^{\text{win}})| \cdot h_i^{D_i}(1-h_i)^{1-D_i} \cdot \sum_{j=1}^{K} p_j \, s(\sigma^*(h_i),\, r_{2,j}^*) \cdot f(h_i \mid x_j)} \tag{11}$$

where $h_i = \tau(r_i^{\text{win}})$. This is the probability that: a borrower with type $h_i$ is charged rate $\sigma^*(h_i)$ (captured by the Jacobian $|\tau'|$), stays with bank 1 (the sum weights each competitive environment by the joint density $p_j \cdot f(h_i|x_j)$, reflecting both the population frequency of $x_j$ and how likely type $h_i$ is given $x_j$), and defaults or not (the Bernoulli term).

*Remark* 10. Alternatively, the selection sum can be written as $f(h_i) \cdot \sum_j \pi_j(h_i)\, s(\sigma^*(h_i), r_{2,j}^*)$ where $f(h) = \sum_k p_k f(h|x_k)$ is the marginal density. This form makes explicit that bank 1 weights the competitive environments by the <u>posterior</u> $\pi_j(h)$, not the unconditional $p_j$.

*Remark* 11 (Simplification relative to the public-$\mathbf{x}$ case). When $\mathbf{x}$ is public information, the bank-1 likelihood involves $\tau(r_1; r_2)$—the type recovery depends on the latent $r_2$, so $h$ varies across the integration. Here, $h_i = \tau(r_i^{\text{win}})$ is <u>known</u> and constant across the summation. The sum over $j$ affects only the selection probability, not the identity of the borrower's type.

## 4.2 Bank-2 Winners ($W_i = 2$)

For a bank-2 winner, $r_i^{\text{win}} = r_{2,j_i}^*$ for some $j_i \in \{1, \ldots, K\}$, observed directly. The type $h_i$ is latent:

$$\boxed{\mathcal{L}_i^{(2)}(\theta) = p_{j_i} \cdot \int_{\underline{h}}^{\bar{h}} \big(1 - s(\sigma^*(h), r_{2,j_i}^*)\big) \cdot h^{D_i}(1-h)^{1-D_i} \cdot f(h \mid x_{j_i})\, dh} \tag{12}$$

This has the same form as in the public-$\mathbf{x}$ case, except that $\sigma^*(h)$ no longer depends on $r_2$ (it is a function of $h$ alone).

## 4.3 Full Log-Likelihood

$$\ell(\theta) = \sum_{i=1}^{N} \Big[ \mathbf{1}(W_i = 1) \ln \mathcal{L}_i^{(1)}(\theta) + \mathbf{1}(W_i = 2) \ln \mathcal{L}_i^{(2)}(\theta) \Big] \tag{13}$$

## 4.4 Computational Strategy

1. Pre-compute $\sigma^*(h)$ on a fine grid of $h$ by solving the fixed-point equation (6). This requires summing over the $K$ support points at each iteration—no numerical integration over $G$ is needed.

2. Construct $\tau(r_1)$ by numerical inversion.

3. For bank-1 winners: $h_i = \tau(r_i^{\text{win}})$ is computed once; the sum over $j$ in (11) involves $K$ logit evaluations.

4. For bank-2 winners: the integral over $h$ in (12) is evaluated by quadrature on the $h$ grid.

5. The equilibrium $\sigma^*(h)$ depends on $(\alpha, \lambda, r_{2,1}^*, \ldots, r_{2,K}^*, p_1, \ldots, p_K, f(\cdot|x_j))$ through the posterior weights $\pi_j(h)$. Since $r_{2,j}^*$ are observed, the equilibrium effectively depends on $(\alpha, \lambda, p_1, \ldots, p_K, f(\cdot|x_j))$ and must be recomputed at each parameter evaluation. In practice, one parametrizes $f(h|x_j)$ with a flexible family (e.g., Beta, sieve basis) for computational tractability.

# 5 Comparison with the Public-x Case

## 5.1 Identification Comparison

| | **Public x, observed** (both banks know $\mathbf{x}$, econometrician sees $\mathbf{x}$) | **Public x, unobserved** (both banks know $\mathbf{x}$, econometrician doesn't) | **Private x** (only bank 2 knows $\mathbf{x}$) |
|---|---|---|---|
| $\sigma^*$ depends on | $(h, \mathbf{x})$ | $(h, \mathbf{x})$ | $h$ only |
| Type recovery | Exact within $\mathbf{x}$ | Fails (mixture) | **Exact** |
| $(\alpha, \lambda)$ | Nonparametric | Parametric $F$, $G$ needed | **Nonparametric** |
| $F(h\|\cdot)$ | Nonparametric | Parametric | **Nonpar.** ($K = 2$); Parametric ($K \geq 3$) |
| $G$ | Observed ($\mathbf{x}$ seen) | Parametric | Finite-dim. (Assumption 1) |
| Key bottleneck | Need $\mathbf{x}$ observed | Everything coupled | $(\alpha, \lambda, f(h\|x_j), p_j)$ coupled but overidentified |

## 5.2 Testable Implications: Distinguishing the Models

The two information structures generate different testable predictions.

**Proposition 12** (Distinguishing test)**.** *Define* $V(r_1) \equiv \underline{Var}(D_i \mid r_i^{win} = r_1,\, W_i = 1)$. *Under the private-*$\mathbf{x}$ *information structure:*

$$V(r_1) = \tau(r_1)\big(1 - \tau(r_1)\big) \qquad \textit{(pure Bernoulli variance)}$$

*Under the public-*$\mathbf{x}$ *information structure (econometrician without* $\mathbf{x}$*):*

$$V(r_1) > E[h|r_1, W{=}1]\big(1 - E[h|r_1, W{=}1]\big) \qquad \textit{(excess variance from type mixture)}$$

*whenever more than one type produces rate* $r_1$.

*Proof.* Under the private-$\mathbf{x}$ structure, all bank-1 winners at rate $r_1$ have the same type $h = \tau(r_1)$, so $D_i \sim \text{Bernoulli}(h)$ and $\text{Var}(D_i \mid r_1, W_i{=}1) = h(1-h)$.

Under the public-$\mathbf{x}$ structure, bank-1 winners at rate $r_1$ can have different types (from different $\mathbf{x}$). Let $h$ be random conditional on $(r_1, W{=}1)$. By the law of total variance:

$$\begin{aligned} V(r_1) &= E[\text{Var}(D|h)] + \text{Var}(E[D|h]) \\ &= E[h(1-h) \mid r_1, W{=}1] + \text{Var}(h \mid r_1, W{=}1) \end{aligned}$$

The second term is strictly positive when the conditional distribution of $h$ given $(r_1, W = 1)$ is non-degenerate. $\square$

*Remark* 13 (Practical implementation). To test this, estimate $\hat{h}(r_1) = \hat{E}[D_i \mid r_1, W_i = 1]$ and $\hat{V}(r_1) = \widehat{\mathrm{Var}}(D_i \mid r_1, W_i = 1)$ nonparametrically (e.g., kernel regression). Under the private-$\mathbf{x}$ structure, pooling borrowers in a narrow bin around $r_1$ should yield $\hat{V} \approx \hat{h}(1 - \hat{h})$. Under the public-$\mathbf{x}$ structure, $\hat{V}$ should systematically exceed $\hat{h}(1 - \hat{h})$. A formal test can be based on the difference $\hat{V}(r_1) - \hat{h}(r_1)(1 - \hat{h}(r_1))$, which should be zero under the private-$\mathbf{x}$ structure.

# A Identification with Continuous Covariates

This appendix extends the identification analysis to the case where the covariates $\mathbf{x}$ take a continuum of values. When $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d$ with a continuous distribution, bank 2's equilibrium strategy $r_2^*(\mathbf{x})$ induces a continuous distribution $G$ of entrant rates. The distribution $G$ is now infinite-dimensional, which creates an additional identification challenge.

## A.1 The Infinite-Dimensional Problem

The bank-1 FOC (5) continues to hold, now with the posterior density $\pi(r_2|h) = g(r_2)f(h|r_2)/f(h)$:

$$\alpha \cdot m(h) = \frac{\int s(\sigma^*(h), r_2)\, \pi(r_2|h)\, dr_2}{\int s(\sigma^*(h), r_2)(1 - s(\sigma^*(h), r_2))\, \pi(r_2|h)\, dr_2} \qquad \text{for all } h$$

This is a continuum of equations (one per $h$) in the unknowns $(\alpha, \lambda, G, \theta_F)$. Since both $G$ and $\pi(\cdot|h)$ are infinite-dimensional, the system cannot be solved without restricting $G$.

If $G$ were known (e.g., degenerate at a single $r_2$), the system would reduce to the standard identifying restriction:

$$\ln(\alpha\, m(h) - 1) = \lambda - \alpha\big(\sigma^*(h) - r_2\big)$$

which identifies $(\alpha, \lambda)$ from two values of $h$. With a non-degenerate continuous $G$, this simplification is unavailable.

Nonetheless, the FOC provides a strong restriction. A continuum of equations (indexed by $h$) must simultaneously hold, which is far more restrictive than a system with finitely many equations.

## A.2 Parametric Approach

A parametric assumption on $G$ restores tractability.

*Assumption* 14 (Parametric mixing distribution). (i) **Type distribution.** $h \mid r_2 \sim F(h;\, \theta_F(r_2))$, where $F$ belongs to a known parametric family and $\theta_F(r_2)$ varies with $r_2$ through a finite-dimensional parameterization.

**Example:** $h \mid r_2 \sim \text{Beta}(a, b_0 + b_1 r_2)$ on $[\underline{h}, \bar{h}]$, giving $\theta_F = (a, b_0, b_1)$.

(ii) **Mixing distribution.** $r_2 \sim G(r_2; \phi)$ belongs to a known parametric family.

**Example:** $r_2 \sim \text{LogNormal}(\mu_r, \sigma_r^2)$, giving $\phi = (\mu_r, \sigma_r)$.

The full parameter vector is $\theta = (\alpha, \lambda, \theta_F, \phi)$. Identification proceeds as in the main text (Stages 1–4), with two modifications:

- Stage 2 of the main text (observing support points) is replaced by a parametric assumption on $G(\cdot; \phi)$. The FOC system (**??**) now constrains $(\alpha, \lambda, \theta_F, \phi)$ jointly, with the integrals computed under the posterior $\pi(r_2|h; \theta_F, \phi)$.
- Stage 3 (joint identification) identifies $(\alpha, \lambda, \theta_F, \phi)$ from the FOC plus bank-2 default rates, since the total dimension is finite and the FOC holds for a continuum of $h$. For the LogNormal example, $\dim(\alpha, \lambda, \theta_F, \mu_r, \sigma_r)$ is small relative to the continuum of FOC restrictions.
- The overidentifying restriction on $\phi$ from the selected distribution of bank-2 rates (Bayes' rule inversion of $g_{r_2|W=2}$) provides a separate check.

## A.3 Comparison

The finite-$\mathbf{x}$ assumption (main text) is strictly weaker: no parametric assumption on $G$ is needed (the support points $r^*_{2,j}$ are directly observed), and for $K = 2$ the conditional densities $f(h|x_j)$ are nonparametrically identified. For $K \geq 3$, a parametric family for $f(h|x_j)$ is needed (the pointwise system is underdetermined), but the system remains massively overidentified. The parametric specifications of Assumption 14 are needed for the continuous-$\mathbf{x}$ case, where both $G$ and the posterior become infinite-dimensional. In practice, since credit covariates are typically reported in discrete categories (credit score bins, income brackets, geographic regions), the finite-$\mathbf{x}$ approach applies naturally.

The likelihood under continuous $\mathbf{x}$ is:

$$\mathcal{L}_i^{(1)}(\theta) = |\tau'(r_i^{\text{win}})| \cdot h_i^{D_i}(1-h_i)^{1-D_i} \cdot \int s(\sigma^*(h_i), r_2) \cdot f(h_i \mid r_2; \theta_F) \cdot g(r_2; \phi)\, dr_2 \tag{14}$$

$$\mathcal{L}_i^{(2)}(\theta) = g(r_i^{\text{win}}; \phi) \cdot \int_{\underline{h}}^{\bar{h}} (1 - s(\sigma^*(h), r_i^{\text{win}})) \cdot h^{D_i}(1-h)^{1-D_i} \cdot f(h \mid r_i^{\text{win}}; \theta_F)\, dh \tag{15}$$

where the bank-1 likelihood now requires quadrature over $r_2$ under $g(\cdot; \phi)$, and the bank-2 likelihood uses $g(r_i^{\text{win}}; \phi)$ as the marginal density of the entrant rate.